

Mapping Africa's Endangered Archaeological Sites and Monuments (MAEASaM): Grid

Arches Resource Model Graph and Reference Data (Thesaurus)

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Schema Attributes and Definitions

All resource model metadata templates in the MAEASaM database, including Remote Sensing Resource Model, are described using the following shared attributes:

Group Name: In Arches, this corresponds to the Semantic Node. It represents the parent category under which related metadata elements are organised.

Node Name: Also referred to as a metadata element. This is the primary descriptive field used to document a Remote Sensing record.

CIDOC CRM Class: Indicates the equivalent class in the CIDOC CRM ontology. This mapping supports semantic interoperability.

CIDOC CRM Property: Indicates the equivalent property in CIDOC CRM ontology. This mapping supports semantic interoperability.

Connector Node: Specifies whether the node establishes a relationship with another resource model. Values are “yes” or “no”.

Connected To: If the Connector Node value is “yes”, this field specifies the name of the related Resource Model.

Data Type: Defines the type of data entered in Arches. Common data types include: string, date, concept list, and resource-instance-list.

Necessity: Indicates whether the node is required when describing a record. Possible values are:

“mandatory”

“strongly suggested”

Cardinality: Specifies how many values a node can contain. In the MAEASaM Arches context, four options are used:

1 - Only one value allowed

0..1 - Optional, maximum one value

1..n - At least one value required, multiple allowed

0..n - Optional, multiple values allowed

Scope Note: Provides a definition and conceptual explanation of the node.

Protocol for Entry: Explains how information should be entered into this node, including formatting rules, conventions, or controlled vocabulary guidance

Section I: Overview

I.1 Objectives and Scope

The objective of the grid model is to create a uniform grid structure that can be used throughout all of the project countries. This is required because no standardised grid system for all the project countries currently exists. Each grid square size matches that used by the Endangered Archaeology in the Middle East and North Africa (EAMENA) grid (<https://eamena.org/advanced-use#rm-gs>), with each grid covering a geographic extent of 15 x 15 degrees (longitude and latitude) and is approximately 40 x 40 km (1 600 km²). Designing the MAEASaM grid to be compatible with the EAMENA grid enables integration into a broader regional grid system spanning Africa and the Middle East. Each grid square corresponds to the geographic extent of a 1:50 000 topographic map, which allows for the integration of legacy historical maps housed at the project partner institutions' repositories directly into the MAEASaM grid. The Grid Model provides a standardised system to allow for the integration of site data identified via remote sensing techniques and through records-based research - legacy site data, and is used as a reference source in the Site, Remote sensing and Information (will later be replaced by the Maps) resource models.

Geographical extent: The design of the grid was based on the territorial boundaries of the MAEASaM project's 16 partner African countries, as well as a limited extent of the surrounding territory of each country. Therefore, each country grid covers the country's territorial boundaries and relevant water bodies, such as the large lakes of Eastern Africa. Grids located in coastal countries were further extended into the territorial waters of each country to allow for the future integration of maritime cultural heritage.

I.2 History of the creation of the MAEASaM Grid Resource Model

In order to streamline the creation of the grid and to ensure continuity, a Python script (Appendix B) was designed that created the grid for the extent of Africa, both for level 1 and for level 2 (each level 1 grid is subdivided into 9 smaller grids which make up the level 2 grid). The name of each grid was determined by the X and Y coordinates of the centroid of each grid, with a precision of 3 decimal points. The script additionally contains a section that also provides a shorter version of this name structure which reflects the centroid rounded down to one precision. Appendix A contains a QGIS expression that can be used to create the longer grid name (this expression was incorporated into the Python script). Once the grid was created with a geopackage as the output file, the geopackage was imported into QGIS. An admin level 0 layer (consisting of the boundaries of the project partner countries obtained from the Humanitarian Data Exchange) was used to select the grids from the Africa grid. These were then exported as a GeoJSON file. Once this was done, the following expression was used to add the MAEASaM Grid ID to the grid file:

```
'GRID-' || lpad(@row_number, 7, '0')
```

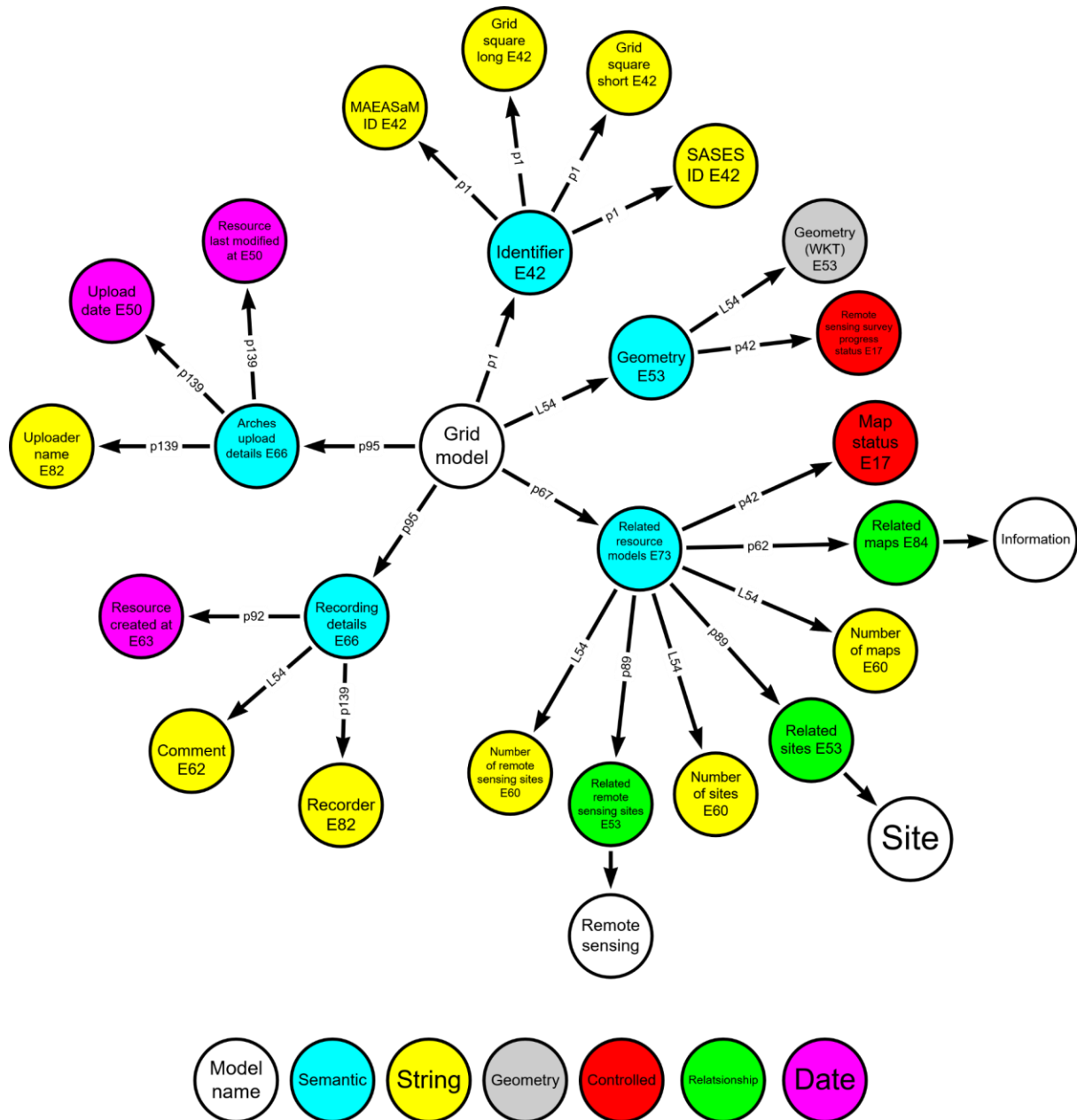
The above QGIS expression will generate an ID with the following structure: GRID-00000001. To prevent duplication, the expression uses the row number for each grid (automatically assigned by QGIS) and then uses that number to generate the MAEASaM ID. The sequential numbering will remove a zero for 10, 100 and 1000 intervals. By default, numbering in QGIS starts from 1 for the first row (this differs from Python counting which starts from 0 and should be compensated for if Python scripting is to be used). When a data set is loaded into QGIS, the organisation of the rows can vary, and thus this expression will not provide the same numbering if the row order is different. This created a MAEASaM ID for each grid square.

Some of the partner countries use a grid system called the Standardized Site Enumeration System (SASES) developed by Charles M Nelson (Nelson, 1993). This is a grid-based system that divides the African continent into a series of grids that have a geographic extent equal to that of the MAEASaM project. Therefore, to allow for the full integration of the SASES grid system into the project's grid, each grid square in the MAEASaM grid was also assigned a SASES ID based on the design parameters provided by Nelson (1993). The parameters were integrated into the grid Python creation script (appendix B) and were assigned at grid creation.

With the final determination of the grid geographic extent, as well as the assigning of the MAEASaM and SASES IDs, the final version of the Arches Grid model file was created. After the creation of a grid geopackage that covered every partner country, this grid was then also further subdivided into country-specific geopackages for each of us.

2. Resource model graph

2.1 Structure of the Grid Resource Graph



A graphical representation of the graph structure used in the MAEASaM Grid resource model using the CIDOC-CRM version 6.2 ontology (<https://cidoc-crm.org/Version/version-6.2>). Each colour represents a different data node used within the Arches database, the CIDOC class (E) is used and how that node relates to other nodes via the CIDOC property (p).

2.2 Grid Resource Model Nodes and CIDOC-CRM Structure

Semantic Node: Identifier

Node Name: MAEASaM ID

CIDOC CRM Class: E42 Identifier

CIDOC CRM Property: p1 is identified by

Connector Node: No

Connected to: N/A

Data type: String

Name of concepts collection: N/A

Cardinality: 1

Necessity: Mandatory

Scope note (description of node)

This node captures the value of the MAEASaM ID for each grid square and uses the E42 identifier value in the CIDOC-CRM (v6.2) ontology and is linked with the p1 is identified by property. The ID is an alphanumeric identifier separated into two sections, divided by a dash (-). The first section of the id has the name of the model to which it belongs, and for this model it is represented by "GRID". The numerical part follows sequentially, starting from 1. The design of the ID structure uses a fixed character size, and thus the first number will be preceded by 8 zeros, to create a numerical number consisting of 8 characters. As the sequential number increases, for every 10, 100, 1000 etc, the corresponding zero will be removed to ensure the character length remains 8 characters.

Example:

GRID-00000001

GRID-00000010

GRID-00000100

GRID-00001000

Protocol for entry:

- Allow only one ID per entry.
- Determine the ID value based on the relevant grid square.
- The relevant grid square either:
 - the grid square being worked on that overlaps the scanned map's geographical extent, or
 - the grid square that contains the relevant site within its geographical extent.

Node Name: Grid square long
CIDOC CRM Class: E42 Identifier
CIDOC CRM Property: p1 is identified by
Connector Node: No
Connected to: N/A
Data type: String
Name of concepts collection: N/A
Cardinality: 1
Necessity: Mandatory

Scope note (description of node)

The function of this node is to provide the primary name value for the relevant grid square. An identifier class is used to assign a unique and permanent value within or between organisations. To generate the name value, the coordinates of the central point of each grid square were used. The structure of the name is based on decimal degrees, with the longitude and latitude indicators being compass direction (NWES) with a precision of 3.

Example:

W17.375N14.875

Protocol for entry:

- The ID should be added as a completed ID
- Only one ID is allowed per entry
- The ID value is determined by the affected grid
- The affected grid is either:
 - the grid being worked on that overlaps with the scanned map's geographical extent, or
 - the grid that contains the relevant site within its geographical extent

Node Name: Grid square short
CIDOC CRM Class: E42 Identifier
CIDOC CRM Property: p1 is identified by
Connector Node: No
Connected to: N/A
Data type: String
Name of concepts collection: N/A
Cardinality: 1
Necessity: Mandatory

Scope note (description of node)

The function of this node is to provide the secondary name value for the relevant grid square. An identifier class is used to assign a unique and permanent value within or between organisations. To generate the name value, the coordinates of the central point of each grid

square were used. The structure of the name is based on decimal degrees, with the longitude and latitude indicators being compass direction (NWES) with a precision of 1.

Example:

W17.3N14.8

Protocol for entry:

- The ID should be added as a completed ID
- Only one ID is allowed per entry
- The ID value is determined by the affected grid
- The affected grid is either:
 - the grid being worked on that overlaps with the scanned map's geographical extent, or
 - the grid that contains the relevant site within its geographical extent

Node Name: SASES ID

CIDOC CRM Class: E42 Identifier

CIDOC CRM Property: p1 is identified by

Connector Node: No

Connected to: N/A

Data type: String

Name of concepts collection: N/A

Cardinality: 1

Necessity: Mandatory

Scope note (description of node)

The function of this node is to provide the Standardized Site Enumeration System (SASES) ID for the related grid square (for more details, see Nelson, 1993). The node uses the E42 class with the p1 property. The SASES system was designed to be implementable for the entire African continent and as such every square in MAEASaM ID has an associated SASES ID (the SASES system is also designed to be used for 1:50 000 maps, which allow for incorporation into the MAEASaM grid system). The SASES system is mostly used in Eastern Africa.

Example:

EeAl

Protocol for entry:

- Each grid within the MAEASaM Grid has an associated SASES ID value
- When entering data, the SASES ID value of the associated MAEASaM Grid should be added
- The SASES ID consists of a 4-character alphabetic value
- Two uppercase letters and two lowercase letters

Semantic Node: Geometry

Node Name: Geometry (WKT)
CIDOC CRM Class: E53 Place
CIDOC CRM Property: L54 is same as
Connector Node: No
Connected to: N/A
Data type: String
Name of concepts collection: N/A
Cardinality: 1..n
Necessity: Mandatory

Scope note (description of node)

This node provides information related to the geographical extent of each grid square as it relates to the earth surface. The node uses the E53 class with the L54 (CIDOC CRM 6.2 extension CRMdig v3.2.1) property. Each grid square has four points that make up the square geometry. The geometry is based on the World Geodetic System (WGS) 84 system, with the longitude and latitude points being presented in Well-known Text (WKT) format. Within the Arches system, geometry is stored as a GeoJSON-feature-collection, with upload formats being Shapefile (SHP), Keyhole Markup Language (KML) and the Geometry JavaScript Object Notation (geoJSON) format. Once uploaded to Arches, all geometry is transformed into geoJSON format.

Example

POLYGON ((-7 25.5,-6.75 25.5,-6.75 25.25,-7 25.25,-7 25.5))

Protocol for entry:

- The arches database reads geometry in WKT (Well-Known Text) format
- The WKT will be a polygon containing the coordinates of the 4 points of the square
- The complete WKT must be added with all brackets included
- The alphabetic prefix (e.g., POLYGON) is part of the WKT format and must be added along with the numeric values

Node Name: Remote sensing survey progress status
CIDOC CRM Class: E17 Type Assignment
CIDOC CRM Property: p42 assigned
Connector Node: No
Connected to: N/A
Data type: Concept-list
Name of concepts collection: Remote sensing progress
Cardinality: 1
Necessity: Mandatory

Scope note (description of node)

The function of this node is to indicate if remote sensing surveying has been undertaken in the specific grid, and uses the E17 class with the p42 property. This is a controlled list node which will indicate if surveying has been started, is in progress or has been completed.

Example:

Surveyed

Protocol for entry:

- This is a controlled vocabulary node
- The value added should match that present in the concept list linked to the node
- Arches uses a case-sensitive matching system requiring a precise match to the concept list value
- Wrong capitalization or additional white spacing will result in an error when uploaded

Semantic Node: Related resource models

Node Name: Map status

CIDOC CRM Class: E17 Type Assignment

CIDOC CRM Property: p42 assigned

Connector Node: No

Connected to: N/A

Data type: Concept-list

Name of concepts collection: Map status

Cardinality: 1..n

Necessity: Mandatory if applicable

Scope note (description of node)

The function of this node is to provide information related to the digitisation status of the map/s associated with the selected grid, and uses the E17 class with the p43 property. Multiple maps may be available for each grid square, and the status of each can be indicated for each related map.

Example:

Digitised

Protocol for entry:

- This is a controlled-list node
- The value must match the value in the Arches collection linked to this node precisely
- Arches uses a case-sensitive matching system
- Wrong capitalization or white spacing will result in an error

Node Name: Related maps

CIDOC CRM Class: E84 Information Carrier

CIDOC CRM Property: p62 depicts

Connector Node: Yes

Connected to: Map resource model

Data type: resource-instance-list

Name of concepts collection: N/A

Cardinality: 0..n

Necessity: Mandatory if applicable

Scope note (description of node)

The function of this node is to provide the information related to the name or identification number of the map in relation to the previous node. This node uses the E84 class with the p62 property. Due to the variation in information provided in maps, with no definitive standardisation being used internationally or historically, this node captures the information used to make reference to the map. This form of identification can be the ID code used or a name given to the map.

Example:

INFO-000000025

Protocol for entry:

- This is a relationship node linked with the Information resource model
- The MAEASaM ID of the map being linked to needs to be added to this value
- For a relationship node value to be uploaded, the resource being linked to must already be uploaded to the database first

Node Name: Number of maps

CIDOC CRM Class: E60 Number

CIDOC CRM Property: L54 same as

Connector Node: No

Connected to: N/A

Data type: String

Name of concepts collection: N/A

Cardinality: 0..n

Necessity: Mandatory if applicable

Scope note (description of node)

The function of this node is to indicate the number of maps that have been digitised, in the process of being digitised or are planned to be digitised that cover the same, or a section, of the geographic extent of the associated grid square. This node uses the E60 class with the L54 (CIDOC CRM 6.2 extension CRMdig v3.2.1) property.

Example:

5

Protocol for entry:

- Provide a number in Western Arabic format

- Indicates the quantity of maps associated with the specific grid

Node Name: Related sites
CIDOC CRM Class: E53 Place
CIDOC CRM Property: p89 contains
Connector Node: Yes
Connected to: Site Resource Model
Data type: Relationship-instance-list
Name of concepts collection: N/A
Cardinality: 0..n
Necessity: Mandatory if applicable

Scope note (description of node)

The function of this node is to provide the information of any site located within the geographic extent of the grid that has been recorded in the national catalogue of the country in which the grid is located or in publication. The node uses the E53 class with the p89 property.

Example:

SITE-TZA-000000100

Protocol for entry:

- This is a relationship node linked with the Site resource model
- The MAEASaM ID of the site being linked to needs to be added to this value
- For a relationship node value to be uploaded, the resource being linked to must already be uploaded to the database first

Node Name: Number of sites
CIDOC CRM Class: E60 Number
CIDOC CRM Property: L54 same as
Connector Node: No
Connected to: N/A
Data type: String
Name of concepts collection: N/A
Cardinality: 0..n
Necessity: Mandatory if applicable

Scope note (description of node)

The function of this node is to record the quantity of archaeological sites that are present in the national registries of the country in which the grid square is located or has been documented within publications. This node uses the E60 class with the L54 (CIDOC CRM 6.2 extension CRMdig v3.2.1) property.

Example:

10

Protocol for entry:

- Provide a number in Western Arabic format
- Indicates the quantity of maps associated with the specific grid.

Node Name: Related remote sensing sites

CIDOC CRM Class: E53 Place

CIDOC CRM Property: p89 contains

Connector Node: Yes

Connected to: Remote Sensing Resource Model

Data type: Relationship-instance-list

Name of concepts collection: N/A

Cardinality: 0..n

Necessity: Mandatory if applicable

Scope note (description of node)

The function of this node is to record the identification value of all the archaeological sites identified via remote sensing within the geographic extent of the grid, and uses the E53 calls with the p89 property. This is a relationship instance node and links to the Remote Sensing resource model.

Example:

RS-ZWE-000000005

Protocol for entry:

- This is a relationship node linked with the Remote Sensing resource model
- The MAEASaM ID of the resource being linked to needs to be added to this value
- For a relationship node value to be uploaded, the resource being linked to must already be uploaded to the database first

Node Name: Number of remote sensing sites

CIDOC CRM Class: E60 Number

CIDOC CRM Property: L54 same as

Connector Node: No

Connected to: N/A

Data type: String

Name of concepts collection: N/A

Cardinality: 0..n

Necessity: If applicable

Scope note (description of node)

The function of this node is to record the quantity of archaeological sites that have been identified via remote sensing survey within the geographical extent of the relevant grid square. This node uses the E60 class with the L54 (CIDOC CRM 6.2 extension CRMdig v3.2.1) property.

Example:

21

Protocol for entry:

- Provide a number in Western Arabic format
- Indicates the quantity of maps associated with the specific grid.

Semantic Node: Recording details

Node Name: Recorder

CIDOC CRM Class: E82 Actor Appellation

CIDOC CRM Property: p139 has alternative form

Connector Node: No

Connected to: N/A

Data type: String

Name of concepts collection: N/A

Cardinality: 1

Necessity: Mandatory

Scope note (description of node)

The function of this node is to record the name of the individual/s that captured the information in the following nodes: Map status, Related maps, Number of maps, Related sites, Number of sites, Remote sensing survey progress status, Related remote sensing sites and Number of remote sensing sites. The node uses the E82 class with the p139 property.

Example:

Renier Hendrik van der Merwe

Protocol for entry:

- The name of the recorder should be the full version of the name.

Node Name: Comment

CIDOC CRM Class: E62 String

CIDOC CRM Property: L54 is same as

Connector Node: No

Connected to: N/A

Data type: String

Name of concepts collection: N/A

Cardinality: 0..n

Necessity: If applicable

Scope note (description of node)

This is a general, all-purpose node that captures any information that is regarded as required for the specific grid, but is not captured in any of the other nodes present within the resource model. It uses the E62 String with the L54 (CIDOC CRM 6.2 extension CRMdig v3.2.1) property.

Example

Remember to avoid special characters as far as possible.

Protocol for entry:

- This is a general text node
- Special characters should be avoided if possible, including: , " ' % # : ;

Node Name: Resource created at

CIDOC CRM Class: E63 Beginning of Existence

CIDOC CRM Property: p92 brought into existence

Connector Node: No

Connected to: N/A

Data type: EDTF

Name of concepts collection: N/A

Cardinality: 1

Necessity: Mandatory

Scope note (description of node)

The function of this node is to record the date at which the associated grid square was created and uses the E62 class with the p92 property.

Example:

2026-01-21

Protocol for entry:

- The date should follow the Gregorian calendar
- Use year, month, and day separated by hyphens (e.g., YYYY-MM-DD)
- Any other format will cause an error during upload

Semantic Node: Arches upload details

Node Name: Uploader name

CIDOC CRM Class: E82 Actor Appellation

CIDOC CRM Property: p139 has alternative form

Connector Node: No

Connected to: N/A

Data type: String

Name of concepts collection: N/A

Cardinality: 1

Necessity: Mandatory

Scope note (description of node)

The function of this node is to capture the information related to the name of the individual who uploaded the information of the associated entry to the Arches database, and uses the E82 class with the p139 prosperity.

Example:

Renier Hendrik van der Merwe

Protocol for entry:

- The name of the individual who uploaded the data to the Arches database should be the full version of the name.

Node Name: Upload date

CIDOC CRM Class: E63 Beginning of Existence

CIDOC CRM Property: p92 brought into existence

Connector Node: No

Connected to: N/A

Data type: EDTF

Name of concepts collection: N/A

Cardinality: 1

Necessity: Mandatory

Scope note (description of node)

The function of this node is to capture the date on which the information of the associated entry was uploaded to the Arches database and uses the E62 class with the p92 property.

Example:

2026-02-19

Protocol for entry:

- The date should follow the Gregorian calendar
- Use year, month, and day separated by hyphens (e.g., YYYY-MM-DD)
- Any other format will cause an error during upload

Node Name: Resource last modified at

CIDOC CRM Class: E63 Beginning of Existence

CIDOC CRM Property: p92 brought into existence

Connector Node: No

Connected to: N/A

Data type: EDTF

Name of concepts collection: N/A

Cardinality: 1

Necessity: Mandatory

Scope note (description of node)

The function of this node is to capture the date on which any modification was made to any nodes related to the associated entry. This node uses the E63 class with the p92 property.

Example:

2026-03-05

Protocol for entry:

- Every time any modification is made to a specific entry within the resource model, the date should be captured as a new value
- All dates already present should be kept and unmodified
- The date should follow the Gregorian calendar
- Use year, month, and day separated by hyphens (e.g., YYYY-MM-DD)
- Any other format will cause an error during upload

Section 3: Comments

References

- Aalber, T., Balze, D., Bekiari, C., Boudouri, L., Crofts, N., Dunsire, G., Eide, Ø., Gill, T., Goerz, G., Hagedorn-Saupe, M., Hiebel, G., Holmen, J., Inkari, J., Iorizzo, D., Kotipelt, J., Kraus, S., Lampe, K.H., Lamsfus, C., Lindenthal, J., Nyman, M., Riva, P., Rold, L., Smiraglia, R., Stein, R., Stiff, M and Žumer, M. 2015. *Definition of the CIDOC Conceptual Reference Mode*. https://cidoc-crm.org/sites/default/files/cidoc_crm_version_6.2.pdf. Access: 2026-03-12
- Nelson, C.M. 1993. A Standardized Site Enumeration System for the Continent of Africa. In *Nyame Akuma* (40). pp. 62-67.

Appendix A

Additional QGIS expressions that can be used to create the grid in QGIS if the Python script is not used.

Square name (long) QGIS expression:

```
CASE WHEN x(centroid($geometry)) >= 0 THEN 'E' ELSE 'W' END ||
CASE WHEN abs(x(centroid($geometry))) < 10 THEN '0' ||
format_number(abs(x(centroid($geometry))), 3)
  ELSE format_number(abs(x(centroid($geometry))), 3)
END ||
CASE WHEN y(centroid($geometry)) >= 0 THEN 'N' ELSE 'S' END ||
CASE WHEN abs(y(centroid($geometry))) < 10 THEN '0' ||
format_number(abs(y(centroid($geometry))), 3)
  ELSE format_number(abs(y(centroid($geometry))), 3)
END
```

Square name (short) expression:

```
CASE WHEN x(centroid($geometry)) >= 0 THEN 'E' ELSE 'W' END ||
CASE WHEN abs(x(centroid($geometry))) < 10 THEN '0' ||
format_number(abs(x(centroid($geometry))), 1)
  ELSE format_number(abs(x(centroid($geometry))), 1)
END ||
CASE WHEN y(centroid($geometry)) >= 0 THEN 'N' ELSE 'S' END ||
CASE WHEN abs(y(centroid($geometry))) < 10 THEN '0' ||
format_number(abs(y(centroid($geometry))), 1)
  ELSE format_number(abs(y(centroid($geometry))), 1)
END
```

Appendix B

Python script used to create the Grid resource model:

```
import geopandas as gpd
from shapely.geometry import Polygon
import pandas as pd

# Define grid parameters
xmin, ymin, xmax, ymax = -25.0, -36.0, 55.0, 38.0
cell_size = 0.25

# Generate grid cells
grid_cells = []
fid = 0

y = ymax
while y > ymin:
    x = xmin
    while x < xmax:
        # Create polygon for each cell
        polygon = Polygon([
            (x, y),
            (x + cell_size, y),
            (x + cell_size, y - cell_size),
            (x, y - cell_size),
            (x, y)
        ])

        # Calculate center coordinates
        center_x = x + cell_size / 2
        center_y = y - cell_size / 2

        # Format GRID_ID with E/W and N/S prefixes
        ew = 'E' if center_x >= 0 else 'W'
        ns = 'N' if center_y >= 0 else 'S'

        abs_x = abs(center_x)
        abs_y = abs(center_y)

        # GRID_ID with 3 decimals and leading zeros
        if abs_x < 10:
            x_str = f'0{abs_x:.3f}'
        else:
            x_str = f'{abs_x:.3f}'

        if abs_y < 10:
            y_str = f'0{abs_y:.3f}'
        else:
            y_str = f'{abs_y:.3f}'

        grid_id = f'{ew}{x_str}{ns}{y_str}'

        # GRID_ID_short with 1 decimal (rounded)
        rounded_x = round(abs_x, 1)
        rounded_y = round(abs_y, 1)
```

```

        if rounded_x < 10:
            x_str_short = f'0{rounded_x:.1f}'
        else:
            x_str_short = f'{rounded_x:.1f}'

        if rounded_y < 10:
            y_str_short = f'0{rounded_y:.1f}'
        else:
            y_str_short = f'{rounded_y:.1f}'

        grid_id_short = f'{ew}{x_str_short}{ns}{y_str_short}'

        grid_cells.append({
            'fid': fid,
            'geometry': polygon,
            'GRID_ID': grid_id,
            'GRID_ID_short': grid_id_short
        })

        fid += 1
        x += cell_size
        y -= cell_size

# Create GeoDataFrame
gdf = gpd.GeoDataFrame(grid_cells, crs='EPSG:4326')

# Set output path to same directory as script
import os
script_dir = os.path.dirname(os.path.abspath(__file__))
output_path = os.path.join(script_dir, 'africa_grid.gpkg')

# Save to file
print(f'Saving to: {output_path}')
gdf.to_file(output_path, driver='GPKG')

print(f'Created {len(gdf)} grid cells')
print(f'Sample GRID_IDs: {gdf["GRID_ID"].head().tolist()}')
print(f'File saved successfully to: {output_path}')

```

Thesaurus